

Elizabeth Rogers

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EDUCATION:

Harvey Mudd College | Claremont, CA

May 2026

B.S. in Mathematics

Relevant Coursework: Reinforcement Learning, Stochastic Operations Research, Network Science, Algorithms, Intermediate Probability, Data Structures & Program Development, Partial Differential Equations

Leadership: Resident Assistant (RA); President, Gender Equity in Math Club; Writing Center Consultant

SKILLS:

Programming Languages: Python, MATLAB, R, Excel, C++, Java

Programming Libraries: NumPy, Pandas, NetworkX, MatPower, Matplotlib, Geopandas, Seaborn

TECHNICAL EXPERIENCE:

Combinatorial Optimization Senior Thesis Project, Harvey Mudd College

August 2025 - May 2026

- Research the NP-Hard network interdiction mixed integer program VIMAX: all-pairs vitality maximization
- Determined bottleneck points of Benders Decomposition approach to solving VIMAX
- Implement speedups to master problem formulation using developed heuristic solver and improve efficiency of subproblem using Pareto-optimal Benders cuts
- Presented thesis results through both oral presentations and poster presentations to faculty, peers, and alumni

Power Systems Researcher, California Institute of Technology

February 2025 - May 2026

- Developed a Health-Aware Optimal Power Flow model under the guidance of Dr. Adam Wierman
- Determined health cost function relating generator pollutant levels and local health impact through linear regression in python and pandas, scikit-learn libraries
- Analyzed health and equity impacts of economic, health-aware, and carbon-aware optimizations through custom simulations in MATPOWER library and data visualization in Matplotlib, GeoPandas, seaborn libraries
- Documented project through writing a detailed project plan, two interim reports, and final report

Algorithm Design and Analysis Researcher, Harvey Mudd College

November 2024 - December 2025

- Formulated iterative projection algorithm for solving large scale, overdetermined, corrupted system of equations
- Developed and executed computational analysis of algorithm convergence using python, numpy, and matplotlib
- Organized proof of algorithm convergence, solved various supporting lemmas in collaborative team of 3

Social Network Dynamics Researcher, Harvey Mudd College

January 2025 - May 2025

- Simulated propagation of fake news through social networks in Python under guidance of Dr. Susan Martonosi
- Revamped existing Python codebase, verifying accuracy of simulations and set up experiments to test potential policy interventions to the spread of fake news
- Wrote comprehensive documentation and tutorials for future users

Quantum Networks Researcher, Institute for Quantum Computing, Univ. of Waterloo

May 2024 - August 2024

- Meticulously designed, planned, and executed quantum optics experiments, characterizing telescope tracking system and camera for the Quantum Encryption and Science Satellite (QEYSSat) under Dr. Thomas Jennewein
- Used knowledge of experimental quantum key distribution to design experiments accurate to lab's specifications
- Optimized image analysis code in MATLAB using vectorization techniques
- Communicated results and significance to overall project in final presentation to twenty person lab group

Computational Physics Researcher, Harvey Mudd College

February 2023 - January 2024

- Investigated material properties of 2-Dimensional lattice crystals with Dr. Sharon J. Gerbode
- Implemented custom image processing code in MATLAB
- Optimized existing analysis code using parallel computing, decreasing runtime by 20 hours to target
- Improved biggest dynamics simulation bottleneck point through vectorization implementation

PUBLICATIONS:

Health-Aware Optimal Power Flow. *Conference paper in preparation.*

Elizabeth L. Rogers, Nicolas Christianson, Christopher Yeh, Shaolei Ren, Adam Wierman.

- Oral Presentation: Student-Faculty Programs Seminar Day, Caltech (August 2025)
- Poster Presentation: Southern California Conference for Undergraduate Research (November 2025)

Hexagonal vortices enable faster colloidal crystal grain coarsening. *Phys. Rev. E; vol 110, iss 1, 014608, 2024*

Helen K. Chaffee, Eric Corona-Oceguera, Chris G. Couto, Avani N. Anne, **Elizabeth L. Rogers**, Aaron L. Galper, Conor M. Floyd, Ananya Venkatachalam, and Sharon J. Gerbode

- Poster Presentation: Conference for Undergraduate Women in Physics (2024)

HONORS AND AWARDS:

Carl F. Braun WAVE Fellow (Caltech | Pasadena, CA)

2025

USEQIP Undergraduate Research Award (Univ. of Waterloo | Ontario, Canada)

2024

ADDITIONAL EXPERIENCE:

Resident Assistant (RA), Harvey Mudd College

August 2025 - May 2026

- Facilitated community building events and served as lead safety contact for 80 person dormitory, fosters inclusive and supportive living environment, handled conflict mediation

Mentorship Coordinator, Gender Equity in Math, Harvey Mudd College

August 2025 - May 2026

- Organized mentorship program, which included the matching of mentors and mentees, securing and allocating monetary support for mentor groups, and planning of community building events such as trivia and craft nights.

On-Air DJ at College Radio

January 2023 - May 2026

- Hosted 1 hour music show at college radio station, KSPC 88.78fm Claremont
- Highlighted underrepresented artists historically excluded from mainstream radio
- Explored electronic, rock, and punk genres and discusses their respective social movements

Writing Center Consultant, Harvey Mudd College

September 2023 - August 2025

- Mentored peers through the academic writing process in six 45-minute sessions per week

President, Minoritized Genders Across Physics, Harvey Mudd College

September 2023 – August 2025

- Organized community building events for 40+ members, focused on fostering welcoming spaces and connection
- Handled administrative side of club, including budget and treasury duties

Tour Guide, Harvey Mudd College

September 2022 - November 2023

- Led groups of 30 on personalized tours, prioritizing honesty around my college experience